Meet 0

October 19, 2026

Discrete

You have 20 minutes to complete 6 problems. There is no penalty for guessing. Answers are only valid if they are written in the below table.

First and last name	

p	Answer	p	Answer
1		4	
2		5	
3		6	

1. Verity always tells the truth and Falsity always lies. A mystery number is given to both Verity and Falsity, and you are given that it is between 1 and 10, inclusive. Verity first says "The mystery number is a positive integer greater than 5"! Falsity then says "The mystery number is less than 8"! Given this, how many possible choices are there for the mystery number?

2. In AP Biology, the last ten test scores that Bob got were 100, 100, 100, 100, 100, 80, 70, 14, 14, and 2, in order. The mean of Bob's test scores is a, the median is b, and the mode is c. What is the mean of a, b, and c?

3. The number 20 262 027 has exactly 4 positive integer factors. What are the last two digits of the sum of these factors?

4. A pirate hides his 20 gold coins in four baskets, labelled 1, 2, 3, and 4. In baskets 1, 2, and 3 the amount of coins is a prime number. The amount of coins in basket 4 is more than the amounts of coins in baskets 1, 2, and 3 individually, but less than the combined amounts of coins in baskets 1, 2, and 3. If no two baskets have the same amount of coins, compute the amount of coins in basket 4.

5. Kai and Darren each roll a fair 6-sided die, and the product of Kai and Darren's rolls is equal to D. Suppose there exists more than one possible value for the sum of Kai and Darren's rolls. Compute the sum of all possible values of D.

   Note: A fair 6*-sided die is a die in which each of the* 6 *numbers* 1*,* 2*,* 3*,* 4*,* 5*, and* 6 *have an equal probability of being rolled.*

6. Minnetonka Math Team has 1 000 students, some of whom are friends with each other. Each student has at most 5 friends, and friendship is mutual.

   Each student starts with 1 dodecahedron. Every day, one student gives a dodecahedron to one of their friends. After some number of days, some student has all 1000 dodecahedrons. Compute the smallest possible number of days passed.